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AgriScan — An Overview for Farmers, Policymakers, and Partners

This note explains what AgriScan is, what problem it tries to solve, and how it could help farming communities. It is written for a general readership—farmers, county or national agriculture officials, university leaders, cooperatives, and anyone who wants the main ideas without technical or financial detail.

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Why this matters

Across East Africa, many farmers lose crops because diseases are spotted too late, expert advice is hard to reach, and markets do not always connect growers cleanly with buyers or with trusted suppliers of inputs. Extension services do vital work, but one officer cannot be everywhere at once. At the same time, advanced farming technology is often built for large commercial farms and is out of reach for smallholders.

AgriScan is a digital platform that brings three things closer to the farmer's everyday work: quick help to understand what might be wrong with a crop, practical guidance in

familiar language, and a clearer path from diagnosis to treatment and trade.

What AgriScan offers

1. Crop checks from a phone photo

Farmers can photograph affected leaves (or healthy leaves they want to check). The system uses artificial intelligence—similar in spirit to how phones recognise faces—to suggest whether the plant looks healthy or matches patterns associated with common problems for crops the farmer selects. It is designed to **support** good judgement, not to replace walking the field or speaking to an extension officer when that is possible.

2. AgriBot — an agricultural assistant

A chat-style assistant answers questions about farming practice, pests, and crop care. It can work in **English and Kiswahili**, so farmers can ask in the language they prefer. Where documents and local knowledge have been added to the system, answers can draw on that material as well as general agronomic principles.

3. A simple marketplace idea

The platform recognises different roles: **farmers**, **buyers**, and **agencies** (for example sellers of fertiliser, seed, or crop protection products). The aim is to reduce the gap between “I think I know what the crop needs” and “I can find the right product or buyer without unnecessary middlemen.” Listings and connections are organised so that advice, inputs, and sales can line up more logically than in a fully informal chain.

Who it is for

Who	How they might use AgriScan
Farmers	Scan crops, ask questions, list produce, and explore connections to buyers or suppliers.

Who	How they might use AgriScan
Buyers	Find farmers and produce more directly.
Agrovets and input suppliers	Present products so farmers can match treatment to need.
Institutions and researchers	Over time, aggregated patterns (such as frequent diagnoses in a region) can inform training and policy—always with appropriate privacy rules where personal data is involved.

Principles we care about

- **Accessibility:** Mobile-first design so a ordinary smartphone is enough for core features.
- **Local relevance:** Crops and conditions in East Africa—especially Kenya—are central to how models and advice are built and tested.
- **Honesty about limits:** AI can be wrong; severe outbreaks and animal health questions still belong with qualified experts. The tool is meant to speed up first steps, not to replace human expertise.
- **Language:** Supporting Kiswahili alongside English recognises how people actually seek help in the field.

How rollout is imagined (without dates tied to money)

Work is organised in broad phases so that quality and trust come before scale:

1. **Stabilise and pilot** — Refine payments where farmers top up for services (for example via M-Pesa), improve the experience on phones, and test with a first group of farmers.
2. **Grow use of the marketplace** — Invite more agrovets and buyers, work with cooperatives and radio or community channels so people hear about the platform through trusted networks.
3. **Explore larger-scale tools** — For big fields and cooperatives, multispectral imaging (including approaches used with drones) may in future feed the same

kind of intelligence farmers already get from leaf photos. That step depends on regulation, training, and partnership—not something every farmer needs on day one.

This order keeps the foundation reliable before adding more ambitious features.

How we think about success

Progress can be judged in practical terms, not only in headlines:

- **Use:** Are farmers, buyers, and suppliers actually using scans, chat, and listings?
- **Quality of diagnoses:** Do suggestions align with what agronomists see when they check the same cases?
- **Marketplace:** Are farmers and buyers reporting useful connections after a scan or after harvest?
- **Farmer experience:** Simple surveys (for example by SMS) can ask whether people feel they treated crops more appropriately or wasted less on the wrong chemicals.

These measures help the team improve the product and help partners see whether the platform is delivering real-world value.

What AgriScan is not

- It is **not** a substitute for national extension policy or for emergency response when a pest or disease is declared.
- It is **not** a guarantee of a loan or a price; any link to credit or risk scoring in technical versions of the platform is separate and requires clear consent where personal data is shared.
- It is **not** only for experts—language and layout aim to include people who are confident on a phone but not in reading long English reports.

Further reading

For technical architecture, data models, and API detail, see SYSTEM_DOCUMENTATION.md. Background and objectives are summarised in proposal.md.

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AgriScan — bridging practical AI, local languages, and agricultural markets for smallholders and semi-commercial growers in East Africa.